No Detector to Rule Them All

ALAN PEROTTI

MARCO NURISSO, Polytechnic of Turin, Turin, TO, Italy

MIRKO ZAFFARONI

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Alan Perotti[*]
CENTAI Institute
Turin, Italy, IT
alan.perotti@centai.eu

Marco Nurisso[*]
Polytechnic Institute of Turin
Turin, Italy, IT
marco.nurisso@centai.eu

Mirko Zaffaroni[*]
CENTAI Institute
Turin, Italy, IT
mirko.zaffaroni@centai.eu

Abstract

Deepfakes—hyper-realistic synthetic images generated by advanced AI models—pose a growing threat to privacy, security, fairness, and trust in digital media. As generative models become more sophisticated, detecting whether an image is real or synthetic is no longer a purely technical challenge, but a critical societal imperative. Robust deepfake detection is essential for preserving the integrity of online information, protecting individuals from identity misuse, and ensuring transparency in AI-generated content.

In this work, we conduct a comprehensive benchmarking study of open-source deepfake detection models, evaluating their generalization capability across diverse datasets and previously unseen synthetic images. Our results reveal a concerning gap: current detectors struggle to maintain consistent performance across different generative models and datasets, even when those models share similar characteristics. This inconsistency highlights a pressing vulnerability in our defenses against misinformation and manipulation.

We also explore the potential of ensemble-based approaches, finding that simple combinations of detectors often outperform individual models. Beyond binary classification, our feature-level analysis uncovers structured and informative representations that point to a deeper interpretability potential in existing systems.

These findings suggest promising directions for future research: instead of solely aggregating classification decisions, models that merge extracted features appear to be the superior approach for achieving state-of-the-art deepfake detection, thus mitigating the societal impact of such content.

CCS Concepts

• Computing methodologies → Computer vision; **• Applied computing → Computer forensics**; **• Security and privacy → Human and societal aspects of security and privacy**.

Keywords

deepfake detection, generalisation

[*]All authors contributed equally to this research.

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1 Introduction

The rapid advancement of synthetic media generation has led to increasingly realistic deepfakes, posing significant challenges to traditional verification methods. The emergence of generative adversarial networks (GANs) [27] revolutionised synthetic image generation, enabling the creation of highly realistic faces and objects. Further refinements came with architectures such as StyleGAN [37, 38] and BigGAN [14], which improved control over style transfer and high-resolution image synthesis. More recently, diffusion models [23, 31, 62] have set new benchmarks in terms of fidelity, providing enhanced realism and greater flexibility in image generation. Given the rapid progress of these techniques, they are likely to continue driving advancements in generative AI. Simultaneously, transformer-based multimodal models [54, 59] are emerging as key players in the evolution of deepfake technology, further broadening the scope of synthetic media applications.

The increasing accessibility of generative models has raised significant concerns over their misuse, particularly in misinformation campaigns, identity fraud, and social engineering [17]. Deepfakes have been exploited in disinformation strategies, making it increasingly difficult to trust online content [25]. In addition, automated deepfake-based social media bots introduce new challenges for detecting coordinated influence operations. The ever-improving quality of generative models underscores the urgent need for robust detection methods capable of adapting to evolving deepfake technologies.

To address these risks, deepfake detection techniques have advanced alongside generative models. Early detection approaches focused on handcrafted features, detecting inconsistencies in image artifacts, illumination, and colour components [15, 42]. However, as deepfake realism improved, these methods became less effective, prompting a shift toward deep learning-based classifiers, particularly CNN-based architectures [46, 74]. More recent efforts have explored frequency-space learning [69], gradient-based techniques [71], and contrastive learning strategies [9, 20] to improve robustness against unseen deepfake techniques. Despite these advancements, deepfake detectors often struggle to generalise across different datasets and novel generative models, as evidenced by studies on cross-dataset performance degradation [26, 72, 73]. The challenge of improving generalisation remains a critical research problem.

In this work, we conduct a comprehensive benchmarking study to systematically evaluate the generalisation capabilities of modern deepfake detection models. Unlike prior studies that assess models in isolated settings, we examine detection performance across multiple datasets and generative techniques, highlighting fundamental limitations in current approaches. Specifically, we analyse the extent of performance degradation when detectors are applied to synthetic images generated by previously unseen models, emphasizing the ongoing difficulty of achieving reliable cross-dataset generalisation.

Furthermore, we investigate ensembling strategies as a means of improving deepfake detection performance. By integrating multiple detectors, we show that ensemble-based approaches enhance robustness and mitigate dataset-specific biases. In addition to conventional ensembling techniques, we explore the potential of latent feature fusion, leveraging the feature representations from different models to improve classification. Our findings suggest that while binary classifiers remain the predominant approach in deepfake detection, feature-based fusion strategies hold substantial promise for future research. Transitioning toward feature-based methods may lead to more adaptable and interpretable detection frameworks, moving beyond simple classification to enable a deeper understanding of synthetic content.

The remainder of this paper is structured as follows. Section 2 provides an overview of the datasets and deepfake generation techniques used in this paper. Section 3 describes the deepfake detection models evaluated. Section 4 details the setup and results of our benchmarking experiments, focusing on generalisation across datasets. In Section 5, we explore advanced detection strategies, including ensembling and feature-based approaches. Finally, Section 6 summarises our findings and outlines potential directions for future research.

2 Datasets and Generators

To ensure a comprehensive evaluation of deepfake detection models, we curated a diverse set of publicly available datasets containing both real and synthetic images. Our selection includes datasets from research papers, open-access repositories, and platforms like Kaggle. We aimed to maximise coverage of different generative models and provide a broad representation of synthetic image sources.

Table 1 presents the datasets used in this paper, highlighting the variety of images they contain. These datasets include synthetic representations of human faces, places, objects, and animals, making them well-suited for evaluating model generalisation. Additionally, image resolutions vary significantly, ranging from 128×128 to 2048×2048, reflecting real-world conditions that may challenge detectors lacking robust preprocessing. For each dataset we specify the publication year, a reference or link, an alias for convenience, and the kind of images it contains - for instance, `artf` is a collection encompassing images across all four macro-categories: *real*, *GAN*, *diffusion*, *other*.

Table 2 lists the 41 generative models responsible for producing synthetic images in these datasets. These models span multiple years and various architectures, including GAN-based and diffusion-based approaches (indicated in the *Family* column). Notably, the dataset features images from widely used commercial generators such as DALL-E, Midjourney, and Adobe Firefly. The inclusion of

Table 1: Deepfake Detection Datasets

Year	Alias	Reference	Real	GAN	Diff.	Other
2015	celeba	Liu et al. [43]	✓			
2015	imdb	Rothe et al. [63]	✓			
2019	100k	Neves et al. [49]	✓	✓		
2019	ffhq	Karras et al. [34]	✓			
2019	140k	[Kaggle] [1]	✓	✓		
2020	cnnd	Wang et al. [74]	✓	✓		
2020	dffd	Dang et al. [21]	✓	✓		
2022	sfhq	Beniaguev et al. [12]		✓		
2023	artf	Rahman et al. [58]	✓	✓	✓	✓
2023	cddb	Li et al. [41]				✓
2023	dff	Song et al. [67]		✓		
2023	hifi	Guo et al. [29]		✓		
2023	regms	Asnani et al. [6]		✓		✓
2023	synth	Bammey et al. [8]			✓	
2023	ufd	Ojha et al. [51]			✓	
2023	agfd	Wang et al. [75]			✓	
2023	hf	Barnisin et al. [11]		✓		
2023	sdf	[GitHub] [2]			✓	
2024	df40	Yan et al. [79]		✓	✓	
2024	sft2i	Beniaguev et al. [13]			✓	
2024	d3	Baraldi et al. [10]			✓	

both academic and industry-grade models ensures a well-rounded assessment of deepfake detection capabilities. Our dataset selection process emphasised diversity in generation techniques. Some models, such as `alae` (Adversarial Latent Autoencoder) [55], map images into structured latent spaces, while others, like `cips` (Conditionally-Independent Pixel Synthesis) [5], generate pixels independently without traditional convolutions. These unique methodologies introduce further complexity, testing the adaptability of deepfake detectors.

We thus collected a wide range of generative techniques and image types, bundling a rigorous benchmark for evaluating detection models under challenging conditions. For visual reference, we provide sampled images in Figure 1. Our goal is to evaluate detection models under challenging conditions, and assess their abilities to generalise across deepfake source.

3 Deepfake Detectors

The rapid evolution of generative models demanded a parallel advancement in deepfake detection methodologies. Early deepfake detection focused on signal analysis and CNN-based approaches, but as generative adversarial networks (GANs) and diffusion models have become more sophisticated, detection techniques have had to adapt to identify subtle artifacts, generalise across datasets, and counter adversarial attacks.

For our benchmark, we selected a heterogeneous set of state-of-the-art deepfake detectors, acting as representatives of the various detection approaches in the literature. In order to replicate a real

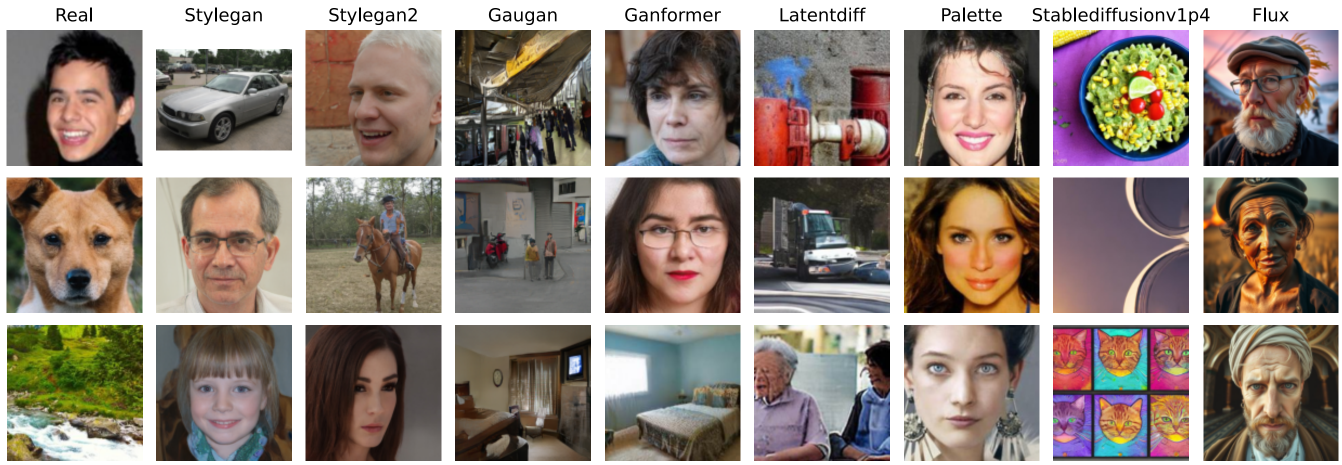

Figure 1: Example of dataset images from various sources.

detection scenario as closely as possible, we chose pre-trained detectors with publicly available code and weights, which can provide a binary real-fake prediction.

The complete list of detectors is shown in Table 3, with the first column assigning aliases to them and the last two columns referring to the broad categories of generative models each detector was trained on.

3.1 Detection of GAN-generated images

The first wave of deepfake detection methods largely relied on image analysis approaches, computing image statistics which could discriminate real and synthetic images. One of the earliest methods, `disp` [42], focuses on detecting inconsistencies in the image's colour components as a means of identifying manipulated images.

Since then, the use of hand-crafted features was for the most part replaced by deep learning, with the task of deepfake detection being approached with deep neural network architectures. Convolutional neural networks (CNNs) trained to identify telltale artifacts in manipulated images proved to be surprisingly good at the identification of GAN-generated images. `cnnd` [74] expanded on this by introducing a ResNet-50 [30] classifier able to generalise well beyond its ProGAN training set.

Texture-based methods also played a crucial role in early detection efforts: `gtex` [44] proposed to employ global texture features, demonstrating that analysing textures can reveal generative patterns that differ from natural images. On a similar note, `pfor` [15] claims that focusing on local image patches instead of the global semantics of the image can provide better detection performance and interpretability.

`gane` [28] critically analysed state-of-the-art GAN detectors, highlighting the importance of pre-training in detectors and the harmful effects of down-sampling and image compression. `otcnn` [46] introduced an orthogonal training strategy for multiple CNNs, reducing overfitting to specific training datasets and improving generalisation.

`lgrad` [71] proposed a method to use gradients as a robust generalised representation of generative artifacts.

More recently, `freq` [69] demonstrated that frequency-aware learning significantly improves generalisability against unseen generative models.

3.2 Detection of Diffusion-generated images

The advent of diffusion models as a generative paradigm rivaling GANs in quality [23], required the development of new detection methods. `diffd` [61] was among the first to establish baseline techniques for identifying diffusion-generated images, including the retraining of previous GAN-based detectors on new datasets. `dire` [76] introduced a model specifically designed for diffusion-based detection. It is able to extract features by mapping images into DDIM's latent space and later re-mapping them into image space.

Additional approaches include spectral analysis-based methods such as `dmimg` [19], which identifies frequency-domain traces left by diffusion models and uses them for classification. `nprd` [70] examined neighbouring pixel relationships as a discriminative feature to spot generated images which went through an up-sampling step. `code` [9] introduced an embedding space built for deepfake detection using contrastive learning, enforcing global-local correspondences through image crops.

3.3 Multimodal and CLIP-Based Detection

Alternative detection strategies have been proposed beyond ad-hoc fake/real classifiers, explicitly to address the issue of generalisation beyond the training dataset.

`ufd` [51] explored the effectiveness of employing feature spaces not learned for the task of deepfake detection, proposing a detector built on top of a pretrained CLIP-ViT [4, 57] model. `clip` [20] expands on this idea, leveraging the features extracted by CLIP to achieve excellent generalisation with a small amount of training samples.

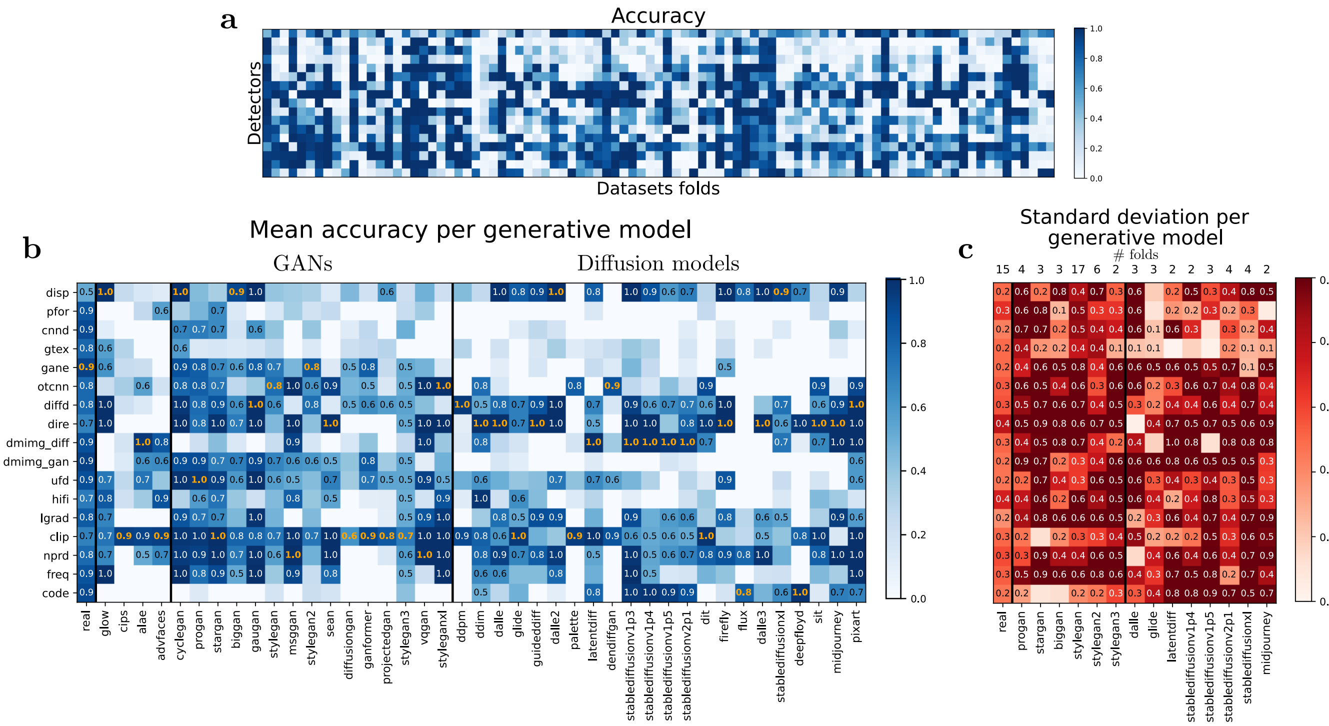

Figure 2: Overview of benchmarking results. (a) Accuracy matrix showing performance of detectors (rows) across data folds (columns). Each cell is colour-coded to represent accuracy, highlighting the lack of a universal detector with consistently high performance. (a) Mean accuracy aggregated by generative model type, separated into GANs and diffusion models, with thick vertical bars separating families of data sources. Results show varying performance trends, with some detectors excelling in specific families but failing to generalise. The best detector performance per generative model is highlighted in orange. (a) Standard deviation across data folds of the accuracy of each generative model, illustrating performance variability.

4 Experimental Benchmarking

4.1 Experimental Setup

The evaluation was conducted using datasets and detectors as detailed in Table 1 and Table 3, respectively. From these datasets, we picked a total of 91 subsets (referred to as data folds) such that each one of them contained either real images or images generated by a *single* generative model in Table 2. When possible, we chose images belonging to the test sets of the datasets employed to minimise the likelihood of testing the detectors on images they were trained on.

To provide a robust comparative framework, additional real image datasets, including COCO, ImageNet, and CelebA, were incorporated as baselines. This inclusion allowed for the assessment of detection models in distinguishing synthetic content from authentic images across diverse visual domains.

For each data fold (e.g., ProGAN-generated images sourced from cnnd), a randomised selection of 1000 images was performed. This resulted in a benchmarking grid consisting of 91 data folds and 17 models, leading to a total of $91 \times 17 \times 1000$ deepfake classification queries.

4.2 Experimental Results

A bird's-eye view of the benchmarking results is presented in Figure 2, panel (a). Each cell represents the accuracy of a specific detector (row) on a given data fold (column). While details are not recognizable from this visualization, the colour-coding of the cells highlights a critical observation: no single detector achieves high performance across all data folds.

Figure 2, panel (b), provides an aggregated representation of the same results by grouping data folds generated by the same model and averaging their accuracies. The thick vertical bars separate the families of generative models (GANs, Diffusion models, other), revealing distinct performance trends. For instance, some detectors, such as gane, perform relatively well on GAN-based datasets but struggle with diffusion-generated images. In general, we see how the first set of detectors, which coincide with the older ones, achieve negligible performances on almost all diffusion models. Conversely, detectors like clip demonstrate better generalisation across both GAN and diffusion families, achieving consistently good accuracy performance in both cases.

Panel (c) in Figure 2, finally, visualises the standard deviation of detector accuracy for each generative model which appears in our dataset in more than one fold. We observe a striking variability across detectors and generative models, even for older GAN-based architectures over which multiple detectors have been trained.

This variability suggests that while some detectors are optimised for specific generative techniques, they fail to maintain robust

Table 2: Synthetic Image Generators

Year	Model	Reference	Family
2017	cyclegan	J. Zhu et al., [80]	GAN
2018	glow	D. Kingma et al., [39]	Diffusion
2018	progan	T. Karras et al., [35]	GAN
2018	stargan	Y. Choi et al., [18]	GAN
2019	biggan	J. Brock et al., [14]	GAN
2019	gaugan	J. Park et al., [53]	GAN
2019	stylegan	T. Karras et al., [37]	GAN
2020	msgggan	A. Karnewar et al., [33]	GAN
2020	advfaces	S. Deb et al., [22]	Diffusion
2020	ddim	J. Song et al., [68]	Diffusion
2020	sean	J. Zhu et al., [81]	GAN
2020	ddpm	J. Ho et al., [31]	Diffusion
2020	alae	I. Pidhorskyi et al., [55]	Other
2020	stylegan 2	T. Karras et al., [38]	GAN
2021	cips	A. Anokhin et al., [5]	Other
2021	diffusiongan	W. Wang et al., [77]	GAN
2021	ganformer	N. Hudson et al., [32]	GAN
2021	projectedgan	A. Sauer et al., [65]	GAN
2021	stylegan 3	T. Karras et al., [36]	GAN
2021	vqgan	P. Esser et al., [24]	GAN
2021	dalle	A. Ramesh et al., [60]	Diffusion
2021	guideddiff	P. Dhariwal et al., [23]	Diffusion
2021	glide	A. Nichol et al., [50]	Diffusion
2022	latentdiff	R. Rombach et al., [62]	Diffusion
2022	stablediffusion v1.3	R. Rombach et al., [62]	Diffusion
2022	stablediffusion v1.4	R. Rombach et al., [62]	Diffusion
2022	stablediffusion v1.5	R. Rombach et al., [62]	Diffusion
2022	stablediffusion v2.1	R. Rombach et al., [62]	Diffusion
2022	palette	A. Saharia et al., [64]	Diffusion
2022	dendiffgan	T. Xiao et al., [78]	Diffusion
2022	styleganxl	A. Sauer et al., [66]	GAN
2022	dalle 2	A. Ramesh et al., [59]	Diffusion
2023	dit	W. Peebles et al., [54]	Diffusion
2023	firefly	Adobe, [3]	Diffusion
2023	stablediffusion xl	R. Podell et al., [56]	Diffusion
2023	dalle 3	OpenAI, [52]	Diffusion
2023	flux	Black Forest Labs, [40]	Diffusion
2024	midjourney 5-6	Midjourney, Inc. [48]	Diffusion
2024	deepfloyd	DeepFloyd Lab, [7]	Diffusion
2024	sit	J. Ma et al., [45]	Diffusion
2024	pixart	X. Chen et al., [16]	Diffusion

performance when applied to diverse datasets. These results collectively highlight the challenges faced by current detection methods in maintaining generalisation and robustness.

Table 3: Deepfake Detection Models

Year	Alias	Reference	GAN	Diff
2018	disp	Li et al. [42]	✓	
2020	cnnd	Wang et al. [74]	✓	
2020	gtex	Liu et al. [44]	✓	
2020	pfor	Chai et al. [15]	✓	
2021	gane	Gragnaniello et al. [28]	✓	
2022	diffd	Ricker et al. [61]	✓	✓
2022	otcnn	Mandelli et al. [46]	✓	
2023	dire	Wang et al. [76]		✓
2023	dmimg	Corvi et al. [19]	✓	✓
2023	hifi	Guo et al. [29]	✓	✓
2023	lgrad	Tan et al. [71]	✓	
2023	ufd	Ojha et al. [51]	✓	
2024	freq	Tan et al. [69]	✓	
2024	nprd	Tan et al. [70]	✓	
2024	clip	Cozzolino et al. [20]		✓
2025	code	Baraldi et al. [9]		✓

5 Towards generalisation

In the previous section, we observed that no single detector is consistently capable of accurately identifying deepfakes originating from diverse sources. This suggests that current detection methods may have limitations in generalising across different types of manipulated content. However, we also noted that certain detectors perform well on specific types of deepfakes, indicating that different models may capture complementary aspects of forgery.

Building on this observation, we hypothesise that integrating multiple detectors could enhance overall detection performance. To explore this idea, we investigate two distinct approaches. The first approach involves aggregation at the verdict level, employing a more conventional ensemble technique to combine the outputs of multiple detectors. It is worth mentioning that [20] conducted a preliminary experiment in this direction.

The second approach focuses on feature-level aggregation, leveraging an exploratory strategy based on dimensionality reduction to identify shared and complementary patterns across different detection models. Through these methods, we aim to improve the robustness and generalisability of deepfake detection systems.

5.1 Model Ensembling

We select a subset of detectors to ensure that all data folds are adequately classified by at least one of them (see Figure 2). To achieve this, we choose the three most recent models: `nprd` [70], `clip` [20], and `code` [9]. These models are selected based on their individual performance across different deepfake categories, ensuring that at least one of them provides reliable predictions for each data fold.

To combine their outputs, we employ two straightforward ensembling strategies:

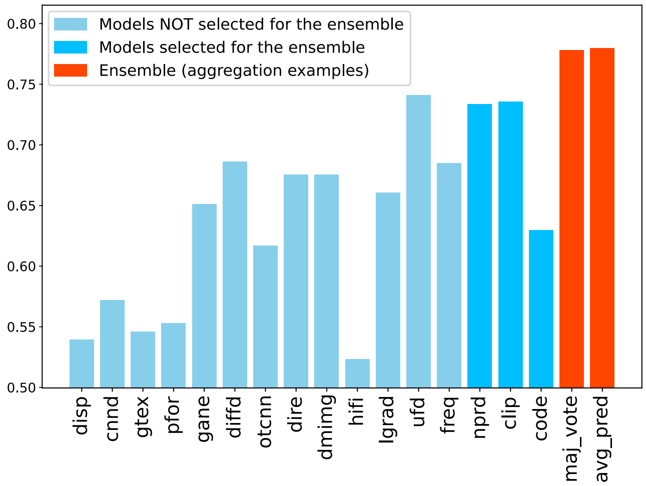

Figure 3: Balanced accuracy of individual detectors and ensemble methods across all data folds of the test set.

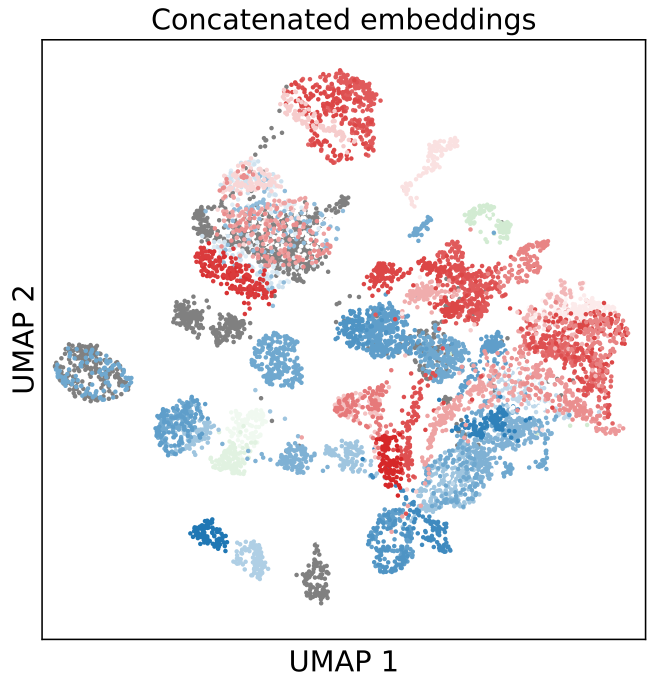

Figure 4: UMAP projection of the concatenated embeddings of `nprd`, `code`, `clip` of a subset of our dataset. Different shades of blue, red and green indicate images belonging to different GAN, Diffusion and other models, respectively. Real images are shown in gray.

- **Majority voting:** We consider the binary verdict from each of the three classifiers and determine the final classification by selecting the most frequently occurring label.
- **Average prediction (avg pred):** Instead of using binary outputs, we take the unthresholded prediction scores from all three models, compute their average, and apply a predefined threshold to obtain the final decision.

To evaluate the effectiveness of these ensemble methods, we conduct experiments on a subset of 100 images from each data fold listed in Table 1. We run all three selected models independently on this dataset and compute the results for both ensembling strategies.

For all cases, we measure the balanced accuracy across real and fake data folds to ensure an unbiased assessment of classification performance. The results of these experiments are presented in Figure 3.

The results in Figure 3 demonstrate that the ensemble methods outperform individual models in terms of balanced accuracy. Models not selected for the ensemble, such as `disp`, `cnnd`, and `gtex`, exhibit poor performance with scores around 0.5, while others like `dire`, `diffd`, and `freq` achieve moderate performance near 0.7. The selected models, `nprd`, `clip`, and `code`, perform consistently well, with balanced accuracies exceeding 0.75. Both ensemble approaches, `maj_vote` and `avg_pred`, further improve performance, achieving nearly identical accuracy scores close to 0.80. This highlights the effectiveness of ensembling in leveraging complementary strengths of the selected models to enhance deepfake detection.

While we adhered to the standard 0.5 threshold commonly used in the literature for consistency (and did not perform any tuning for the ensemble model in general), we experimentally observed that higher thresholds tend to yield better accuracy. Exploring alternative threshold values and calibrating the ensemble in general leaves potential room for improvement in its performance.

5.2 Feature extraction

Since many detectors extract features and then use them to classify images, we wondered whether the quality of the features could be better than the classifier as a whole. In order to explore that, we need to keep the extracted features and only them, de facto using a classifier as a mere feature extractor.

In order to investigate this further, we selected the same three classifiers used in our ensemble approach—`nprd` [70], `clip` [20], and `code` [9]—and extracted their respective feature representations for all test images. Each classifier produces embeddings of different dimensionalities: `code` outputs vectors of size 192, `clip` generates 1024-dimensional embeddings, and `nprd` produces 512-dimensional representations. To construct a unified feature representation, we concatenate these embeddings, resulting in a combined feature vector of size 192+1024+512 = 1728 for each image. To facilitate further analysis, we rescale each part of the concatenated feature vector to ensure consistency in scale across the different detectors' embedding spaces. We then apply Uniform Manifold Approximation and Projection (UMAP) [47] for dimensionality reduction, enabling visualization of the aggregated latent space structure in two dimensions. To examine the underlying structure of the fused feature space, we colour-code the embeddings according to their generative model of origin.

The visualization (Figure 4) reveals a remarkably well-organised feature space, even after the dimensionality reduction step. The separation between different sources suggests that the combined embeddings retain essential information distinguishing generative models. This observation supports the hypothesis that training new classifiers directly on this fused feature space could significantly enhance deepfake detection performance. By leveraging the complementary strengths of different feature extraction models, this approach may lead to more robust and generalisable classifiers capable of effectively distinguishing between real and synthetic images across multiple generative techniques.

6 Conclusions

In this paper, we conducted a comprehensive benchmarking analysis of deepfake detection models, evaluating their ability to generalise across diverse datasets and generative techniques. Our findings underscore a critical challenge at the intersection of generative AI and societal trust: no existing model demonstrates consistent generalisation across the wide spectrum of generative techniques and datasets. This variability presents a tangible risk in real-world applications, where deepfakes can be used maliciously to erode public trust, manipulate narratives, and violate individual privacy.

Our evaluation demonstrated that basic ensemble methods, such as majority voting and average prediction aggregation, can improve classification accuracy compared to individual models. This suggests that leveraging the complementary strengths of different detectors enhances robustness and mitigates dataset-specific biases.

Furthermore, our feature extraction experiments revealed that the latent representations learned by these detectors exhibit well-structured embeddings, with most generative models appearing to be well separated. This indicates significant potential for training classifiers directly on fused feature spaces rather than relying solely on binary classification.

Despite these advancements, generalisation remains a fundamental challenge. Performance degradation across unseen datasets highlights the need for detection models that are more adaptable to emerging generative techniques. Future research should explore hybrid approaches that integrate multimodal analysis, frequency-based techniques, and transformer-based architectures. Additionally, developing methods to better exploit feature representations rather than relying exclusively on classification decisions could provide a promising direction for enhancing deepfake detection.

Ultimately, building efficient deepfake detection frameworks is not only a technical endeavour but a societal necessity; we argue that a shift towards feature-based learning and adaptive, ensemble-driven methodologies is the most promising direction to achieve this goal.

References

[1] [n. d.]. 140k Real and Fake faces. https://www.kaggle.com/datasets/xhlulu/140k-real-and-fake-faces.

[2] [n. d.]. Stable Diffusion Face Dataset. https://github.com/tobecwb/stable-diffusion-face-dataset.

[3] Adobe. 2023. Adobe Firefly.

[4] Dosovitskiy Alexey. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv: 2010.11929* (2020).

[5] Ivan Anokhin, Kirill Demochkin, Taras Khakhulin, Gleb Sterkin, Victor Lempitsky, and Denis Korzhenkov. 2021. Image generators with conditionally-independent pixel synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 14278–14287.

[6] Vishal Asnani, Xi Yin, Tal Hassner, and Xiaoming Liu. 2023. Reverse engineering of generative models: Inferring model hyperparameters from generated images. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2023).

[7] DeepFloyd Lab at StabilityAI. 2023. DeepFloyd IF: a novel state-of-the-art open-source text-to-image model with a high degree of photorealism and language understanding. https://www.deepfloyd.ai/deepfloyd-if.

[8] Quentin Bammey. 2023. Synthbuster: Towards detection of diffusion model generated images. *IEEE Open Journal of Signal Processing* (2023).

[9] Lorenzo Baraldi, Federico Cocchi, Marcella Cornia, Alessandro Nicolosi, and Rita Cucchiara. 2025. Contrasting deepfakes diffusion via contrastive learning and global-local similarities. In *European Conference on Computer Vision*. Springer, 199–216.

[10] Lorenzo Baraldi, Federico Cocchi, Rosario Di Carlo, Marcella Cornia, Lorenzo Baraldi, Alessandro Nicolosi, and Rita Cucchiara. [n. d.]. Diffusion-generated Deepfake Detection dataset (D^3). https://aimagelab.ing.unimore.it/imagelab/page.asp?IdPage=57.

[11] Michal Barnisin. [n. d.]. StyleGAN3 Annotated Images. https://huggingface.co/datasets/balgot/stylegan3-annotated.

[12] David Beniaguev. 2022. Synthetic Faces High Quality (SFHQ) dataset. doi:10.34740/kaggle/dsv/4737549

[13] David Beniaguev. 2024. Synthetic Faces High Quality - Text 2 Image (SFHQ-T2I) Dataset. doi:10.34740/kaggle/dsv/9548853

[14] Andrew Brock, Jeff Donahue, and Karen Simonyan. 2019. Large Scale GAN Training for High Fidelity Natural Image Synthesis. arXiv:1809.11096 [cs.LG] https://arxiv.org/abs/1809.11096

[15] Lucy Chai, David Bau, Ser-Nam Lim, and Phillip Isola. 2020. What makes fake images detectable? understanding properties that generalize. In *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXVI 16*. Springer, 103–120.

[16] Junsong Chen, Jincheng Yu, Chongjian Ge, Lewei Yao, Enze Xie, Yue Wu, Zhongdao Wang, James Kwok, Ping Luo, Huchuan Lu, and Zhenguo Li. 2023. PixArt-α: Fast Training of Diffusion Transformer for Photorealistic Text-to-Image Synthesis. arXiv:2310.00426 [cs.CV]

[17] Robert Chesney and Danielle Keats Citron. 2019. Deep Fakes: A Looming Challenge for Privacy, Democracy, and National Security. *California Law Review* 107, 6 (2019), 1753–1820.

[18] Yunjey Choi, Minje Choi, Munyoung Kim, Jung-Woo Ha, Sunghun Kim, and Jaegul Choo. 2018. Stargan: Unified generative adversarial networks for multi-domain image-to-image translation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 8789–8797.

[19] Riccardo Corvi, Davide Cozzolino, Giada Zingarini, Giovanni Poggi, Koki Nagano, and Luisa Verdoliva. 2023. On the detection of synthetic images generated by diffusion models. In *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 1–5.

[20] Davide Cozzolino, Giovanni Poggi, Riccardo Corvi, Matthias Nießner, and Luisa Verdoliva. 2024. Raising the Bar of AI-generated Image Detection with CLIP. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 4356–4366.

[21] Hao Dang, Feng Liu, Joel Stehouwer, Xiaoming Liu, and Anil K Jain. 2020. On the detection of digital face manipulation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern recognition*. 5781–5790.

[22] Debayan Deb, Jianbang Zhang, and Anil K Jain. 2020. Advfaces: Adversarial face synthesis. In *2020 IEEE International Joint Conference on Biometrics (IJCB)*. IEEE, 1–10.

[23] Prafulla Dhariwal and Alexander Nichol. 2021. Diffusion models beat gans on image synthesis. *Advances in neural information processing systems* 34 (2021), 8780–8794.

[24] Patrick Esser, Robin Rombach, and Bjorn Ommer. 2021. Taming transformers for high-resolution image synthesis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 12873–12883.

[25] Emilio Ferrara. 2020. Deepfake Videos: When Seeing Isn't Believing. *IEEE Spectrum* 57, 8 (2020), 46–51.

[26] Raphael Antonius Frick and Martin Steinebach. 2023. One Detector to Rule Them All? On the Robustness and Generalizability of Current State-of-the-Art Deepfake Detection Methods. In *Proceedings of the 2023 ACM Workshop on Information Hiding and Multimedia Security*. ACM, 1–6.

[27] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. 2014. Generative Adversarial Nets. In *Advances in Neural Information Processing Systems*, Vol. 27. 2672–2680.

[28] Diego Gragnaniello, Davide Cozzolino, Francesco Marra, Giovanni Poggi, and Luisa Verdoliva. 2021. Are GAN generated images easy to detect? A critical analysis of the state-of-the-art. In *2021 IEEE international conference on multimedia and expo (ICME)*. IEEE, 1–6.

[29] Xiao Guo, Xiaohong Liu, Zhiyuan Ren, Steven Grosz, Iacopo Masi, and Xiaoming Liu. 2023. Hierarchical fine-grained image forgery detection and localization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3155–3165.

[30] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer*

vision and pattern recognition. 770–778.
[31] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising diffusion probabilistic models. *Advances in neural information processing systems* 33 (2020), 6840–6851.
[32] Drew A Hudson and Larry Zitnick. 2021. Generative adversarial transformers. In *International conference on machine learning*. PMLR, 4487–4499.
[33] Animesh Karnewar and Oliver Wang. 2020. Msg-gan: Multi-scale gradients for generative adversarial networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 7799–7808.
[34] Tero Karras. 2019. A Style-Based Generator Architecture for Generative Adversarial Networks. *arXiv preprint arXiv:1812.04948* (2019).
[35] Tero Karras, Timo Aila, Samuli Laine, and Jaakko Lehtinen. 2018. Progressive Growing of GANs for Improved Quality, Stability, and Variation. arXiv:1710.10196 [cs.NE] https://arxiv.org/abs/1710.10196
[36] Tero Karras, Miika Aittala, Samuli Laine, Erik Härkönen, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. 2021. Alias-free generative adversarial networks. *Advances in neural information processing systems* 34 (2021), 852–863.
[37] Tero Karras, Samuli Laine, and Timo Aila. 2019. A Style-Based Generator Architecture for Generative Adversarial Networks. arXiv:1812.04948 [cs.NE] https://arxiv.org/abs/1812.04948
[38] Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. 2020. Analyzing and improving the image quality of stylegan. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 8110–8119.
[39] Durk P Kingma and Prafulla Dhariwal. 2018. Glow: Generative flow with invertible 1x1 convolutions. *Advances in neural information processing systems* 31 (2018).
[40] Black Forest Labs. 2023. FLUX. https://github.com/black-forest-labs/flux.
[41] Chuqiao Li, Zhiwu Huang, Danda Pani Paudel, Yabin Wang, Mohamad Shahbazi, Xiaopeng Hong, and Luc Van Gool. 2023. A continual deepfake detection benchmark: Dataset, methods, and essentials. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*. 1339–1349.
[42] H Li, B Li, S Tan, and J Huang. 2018. Detection of deep network generated images using disparities in color components. arXiv 2018. *arXiv preprint arXiv:1808.07276* (2018).
[43] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. 2015. Deep Learning Face Attributes in the Wild. In *Proceedings of International Conference on Computer Vision (ICCV)*.
[44] Zhengzhe Liu, Xiaojuan Qi, and Philip HS Torr. 2020. Global texture enhancement for fake face detection in the wild. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 8060–8069.
[45] Nanye Ma, Mark Goldstein, Michael S. Albergo, Nicholas M. Boffi, Eric Vanden-Eijnden, and Saining Xie. 2024. SiT: Exploring Flow and Diffusion-based Generative Models with Scalable Interpolant Transformers. arXiv:2401.08740 [cs.CV] https://arxiv.org/abs/2401.08740
[46] Sara Mandelli, Nicolò Bonettini, Paolo Bestagini, and Stefano Tubaro. 2022. Detecting gan-generated images by orthogonal training of multiple cnns. In *2022 IEEE International Conference on Image Processing (ICIP)*. IEEE, 3091–3095.
[47] Leland McInnes, John Healy, and James Melville. 2018. Umap: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint arXiv:1802.03426* (2018).
[48] Inc. Mdjourney. 2023. Midjourney. https://www.midjourney.com/home.
[49] João C. Neves, Ruben Tolosana, Ruben Vera-Rodriguez, Vasco Lopes, Hugo Proença, and Julian Fierrez. 2019. GANprintR: Improved Fakes and Evaluation of the State-of-the-Art in Face Manipulation Detection. arXiv:arXiv:1911.05351
[50] Alex Nichol, Prafulla Dhariwal, Aditya Ramesh, Pranav Shyam, Pamela Mishkin, Bob McGrew, Ilya Sutskever, and Mark Chen. 2021. Glide: Towards photorealistic image generation and editing with text-guided diffusion models. *arXiv preprint arXiv:2112.10741* (2021).
[51] Utkarsh Ojha, Yuheng Li, and Yong Jae Lee. 2023. Towards universal fake image detectors that generalize across generative models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 24480–24489.
[52] OpenAI. 2023. DALL·E 3.
[53] Taesung Park, Ming-Yu Liu, Ting-Chun Wang, and Jun-Yan Zhu. 2019. Gaugan: semantic image synthesis with spatially adaptive normalization. In *ACM SIGGRAPH 2019 Real-Time Live!* 1–1.
[54] William Peebles and Saining Xie. 2023. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 4195–4205.
[55] Stanislav Pidhorskyi, Donald A Adjeroh, and Gianfranco Doretto. 2020. Adversarial latent autoencoders. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 14104–14113.
[56] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. 2023. Sdxl: Improving latent diffusion models for high-resolution image synthesis. *arXiv preprint arXiv:2307.01952* (2023).
[57] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In *International conference on machine learning*. PMLR, 8748–8763.
[58] Md Awsafur Rahman, Bishmoy Paul, Najibul Haque Sarker, Zaber Ibn Abdul Hakim, and Shaikh Anowarul Fattah. 2023. Artifact: A large-scale dataset with artificial and factual images for generalizable and robust synthetic image detection. In *2023 IEEE International Conference on Image Processing (ICIP)*. IEEE, 2200–2204.
[59] Aditya Ramesh, Prafulla Dhariwal, Alex Nichol, Casey Chu, and Mark Chen. 2022. Hierarchical text-conditional image generation with clip latents. *arXiv preprint arXiv:2204.06125* 1, 2 (2022), 3.
[60] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. 2021. Zero-shot text-to-image generation. In *International conference on machine learning*. Pmlr, 8821–8831.
[61] Jonas Ricker, Simon Damm, Thorsten Holz, and Asja Fischer. 2022. Towards the detection of diffusion model deepfakes. *arXiv preprint arXiv:2210.14571* (2022).
[62] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 10684–10695.
[63] Rasmus Rothe, Radu Timofte, and Luc Van Gool. 2015. Dex: Deep expectation of apparent age from a single image. In *Proceedings of the IEEE international conference on computer vision workshops*. 10–15.
[64] Chitwan Saharia, William Chan, Huiwen Chang, Chris Lee, Jonathan Ho, Tim Salimans, David Fleet, and Mohammad Norouzi. 2022. Palette: Image-to-image diffusion models. In *ACM SIGGRAPH 2022 conference proceedings*. 1–10.
[65] Axel Sauer, Kashyap Chitta, Jens Müller, and Andreas Geiger. 2021. Projected gans converge faster. *Advances in Neural Information Processing Systems* 34 (2021), 17480–17492.
[66] Axel Sauer, Katja Schwarz, and Andreas Geiger. 2022. Stylegan-xl: Scaling stylegan to large diverse datasets. In *ACM SIGGRAPH 2022 conference proceedings*. 1–10.
[67] Haixu Song, Shiyu Huang, Yinpeng Dong, and Wei-Wei Tu. 2023. Robustness and generalizability of deepfake detection: A study with diffusion models. *arXiv preprint arXiv:2309.02218* (2023).
[68] Jiaming Song, Chenlin Meng, and Stefano Ermon. 2020. Denoising diffusion implicit models. *arXiv preprint arXiv:2010.02502* (2020).
[69] Chuangchuang Tan, Yao Zhao, Shikui Wei, Guanghua Gu, Ping Liu, and Yunchao Wei. 2024. Frequency-Aware Deepfake Detection: Improving Generalizability through Frequency Space Domain Learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 5052–5060.
[70] Chuangchuang Tan, Yao Zhao, Shikui Wei, Guanghua Gu, Ping Liu, and Yunchao Wei. 2024. Rethinking the up-sampling operations in cnn-based generative network for generalizable deepfake detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 28130–28139.
[71] Chuangchuang Tan, Yao Zhao, Shikui Wei, Guanghua Gu, and Yunchao Wei. 2023. Learning on gradients: Generalized artifacts representation for gan-generated images detection. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 12105–12114.
[72] Shahroz Tariq, Sangyup Lee, and Simon S. Woo. 2021. One Detector to Rule Them All: Towards a General Deepfake Attack Detection Framework. In *Proceedings of the Web Conference 2021*. ACM, 3625–3636. doi:10.1145/3442381.3449809
[73] Luisa Verdoliva. 2020. Media Forensics and DeepFakes: An Overview. *IEEE Journal of Selected Topics in Signal Processing* 14, 5 (2020), 910–932.
[74] Sheng-Yu Wang, Oliver Wang, Richard Zhang, Andrew Owens, and Alexei A Efros. 2020. CNN-generated images are surprisingly easy to spot... for now. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 8695–8704.
[75] Zhongqi Wang. 2023. A Generated Face Dataset: AGFD-20K. https://github.com/Robin-WZQ/AGFD-20K.
[76] Zhendong Wang, Jianmin Bao, Wengang Zhou, Weilun Wang, Hezhen Hu, Hong Chen, and Houqiang Li. 2023. Dire for diffusion-generated image detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 22445–22455.
[77] Zhendong Wang, Huangjie Zheng, Pengcheng He, Weizhu Chen, and Mingyuan Zhou. 2022. Diffusion-gan: Training gans with diffusion. *arXiv preprint arXiv:2206.02262* (2022).
[78] Zhisheng Xiao, Karsten Kreis, and Arash Vahdat. 2021. Tackling the generative learning trilemma with denoising diffusion gans. *arXiv preprint arXiv:2112.07804* (2021).
[79] Zhiyuan Yan, Taiping Yao, Shen Chen, Yandan Zhao, Xinghe Fu, Junwei Zhu, Donghao Luo, Chengjie Wang, Shouhong Ding, Yunsheng Wu, et al. 2024. Df40: Toward next-generation deepfake detection. *arXiv preprint arXiv:2406.13495* (2024).
[80] Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A Efros. 2017. Unpaired image-to-image translation using cycle-consistent adversarial networks. In *Proceedings of the IEEE international conference on computer vision*. 2223–2232.
[81] Peihao Zhu, Rameen Abdal, Yipeng Qin, and Peter Wonka. 2020. Sean: Image synthesis with semantic region-adaptive normalization. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 5104–5113.